

Better Executives Work at Better Firms: Identifying Talent Sorting from Mobility Networks

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DRAFT — venue and date TBD

Motivation

Does the allocation of executive talent matter for GDP?

Misallocation lowers aggregate productivity (Hsieh and Klenow 2009)

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Executives are few, but their decisions scale with the firm.

Who runs which firm is a macro question.

A simple assignment model

Firm i employs labor L under fixed factors: fundamentals A_i and executive talent Z_m ,

$$Y_{im} = A_i^{1-\nu} Z_m^\nu \cdot f(L)$$

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Diminishing returns to labor pin down firm scale. Substituting out labor, revenue is Cobb–Douglas in the *two fixed factors*:

$$Y_{im} = A_i^{1-\nu} Z_m^\nu$$

Complementarity means assignment matters

$$\frac{\partial^2 Y}{\partial A \partial Z} > 0$$

The efficient allocation matches better executives to better firms (Becker 1973, Lucas 1978)

Sorting shows up directly in GDP

Sum over the n matches. With $a = \ln A$, $z = \ln Z$ jointly normal,

$$\frac{Y}{n} = \exp \left\{ (1 - \nu)\mu_a + \nu\mu_z + \frac{1}{2}[(1 - \nu)^2\sigma_a^2 + \nu^2\sigma_z^2] + \nu(1 - \nu) \text{Cov}(a, z) \right\}$$

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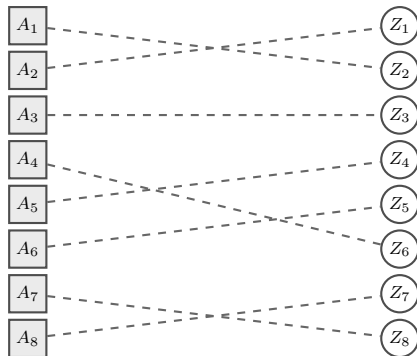
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Holding marginal distributions fixed, GDP increases in $\text{Cov}(a, z)$.

Talent allocation in one picture

firms, ranked

executives, ranked



How far is the economy from the diagonal?

The question, in two layers

Layer 1: a macro question

How strongly are productive firms matched with talented executives?

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How strongly are productive firms matched with talented executives?

Layer 2: measuring something with nothing

Neither A_i nor Z_m is observable.

How can we measure executive talent?

Surveys and diaries of what executives do (Bandiera et al 2020)

Observable characteristics: education, experience, style

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From outcomes: executives who move reveal their persistent contribution (Bertrand and Schoar 2003, Abowd, Kramarz and Margolis 1999)

What we do

- 1 Use the **mobility network**: chains of moves connect firms and executives that never meet
- 2 We do *not* estimate a latent talent for every executive
- 3 We estimate the parameters of a **sorting model**

Our object of interest: four parameters

(A_i, Z_m) of matched pairs jointly lognormal, plus match noise:

$$y_{im} = a_i + z_m + \varepsilon_{im}, \quad \begin{pmatrix} a_i \\ z_m \end{pmatrix} \sim \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} \sigma_a^2 & \rho\sigma_a\sigma_z \\ \rho\sigma_a\sigma_z & \sigma_z^2 \end{bmatrix} \right)$$

$$\boldsymbol{\theta} = (\sigma_a, \quad \sigma_z, \quad \rho, \quad \sigma_\varepsilon)$$

ρ = sorting. How we estimate it: hold that thought.

Prior art: fixed effects

Mobility reveals talent

A firm switches executives; performance jumps.

An executive improves *every* firm they run.

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Treat a_i , z_m as *parameters* and run OLS with two sets of dummies:

$$y_{im} = a_i + z_m + \varepsilon_{im}$$

Labor: (Abowd, Kramarz and Margolis 1999; Andrews et al 2008; Kline, Saggio and Sølvesten 2020).

Executives: (Bertrand and Schoar 2003; Fenizia 2022; Metcalfe, Sollaci and Syverson 2023)

Problem 1: fixed effects need a connected component

Effects are identified only *within* a connected component of the mobility graph.

Standard practice: keep the largest component, drop the rest.

Problem 2: noise biases second moments

Because of ε — not lack of mobility — estimated effects are noisy.

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Second moments, exactly what we care about, are biased: $\widehat{\text{Var}}$ too big, $\widehat{\text{Cov}}$ too small
(Andrews et al 2008; Koren, Orbán and Telegdy 2025)

Limited mobility makes both problems worse.

Both problems are worse for executives

Executive mobility is rare: the graph is very sparse.

Hundreds of thousands of disconnected components.

Firm outcomes are noisy.

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Executive mobility is rare: the graph is very sparse.

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Firm outcomes are noisy.

Estimates are especially noisy — *and* the usual corrections do not apply.

The leave-one-out correction and leverage

Idea (Kline, Saggio and Sølvesten 2020): estimate each effect *without* observation i .

FE estimates are linear in outcomes; observation i enters its own fitted value with weight $h_i = \mathbf{leverage}$. Bias correction re-weights by $1/(1 - h_i)$.

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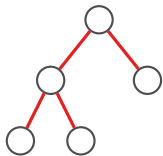
The catch

When $h_i = 1$, the effect is estimable *only* from observation i . The correction divides by zero.

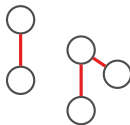
$h_i = 1$ **exactly when edge i is a bridge.**

A detour: trees, forests, cycles, bridges

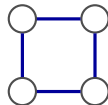
a tree:
no cycles



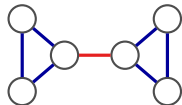
a forest:
disjoint trees



a cycle:
two paths between nodes

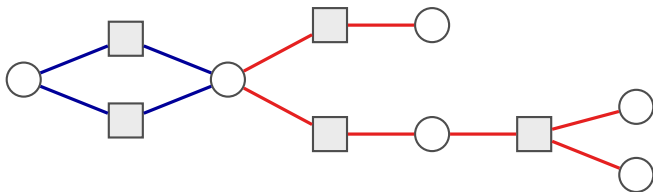


a bridge:
removing it disconnects



On a tree, every edge is a bridge. Bridges also connect cycles: the tree-like edges of a general graph.

Which edges can the leave-out correction use?



blue: on a cycle, leverage < 1 — usable

red: bridges, leverage $= 1$ — dropped

Squares are firms, circles are executives, edges are jobs.

The Hungarian executive mobility network

Universe of Hungarian corporations and their chief executives, 1990–2022. An *observation* is an edge with an outcome: firm performance under that executive.

	Value
Executives	1,304,580
Firms	1,036,231
Firm–executive edges	1,937,552
Connected components	514,085
Bridge share of edges	81.8%
Mean shortest path between executives (giant)	19.5

The graph is close to a forest, with long, sparse chains.

Leave-out estimation keeps one in ten observations

Stage	Matched pairs	%
All unique firm–executive pairs	1,732,154	100.0%
Drop bridges (leverage = 1)	265,995	15.4%
Keep largest leave-out-connected block	175,661	10.1%

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The survivors are highly mobile executives at highly connected firms — not representative.

For a macro question, dropping 90% of the economy will not do.

Our approach: a random-effects model on the network

We model the joint distribution of all outcomes

Instead of estimating 2.3 million fixed effects, model the **joint distribution** of 1.7 million outcomes:

$$\mathbf{x} = (a_1, \dots, a_{N_f}, z_1, \dots, z_{N_m})^\top \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{-1}), \quad y_{im} = a_i + z_m + \varepsilon_{im}$$

A **Gaussian Markov random field** (Rue and Held 2005): the sparse precision matrix $\mathbf{Q}$ encodes the mobility graph.

Same four parameters: $\sigma_a, \sigma_z, \rho, \sigma_\varepsilon$.

Maximum likelihood

$$\ell(\boldsymbol{\theta}) = -\frac{K}{2} \ln(2\pi) - \frac{1}{2} \ln \det \boldsymbol{\Omega} - \frac{1}{2} \mathbf{y}^\top \boldsymbol{\Omega}^{-1} \mathbf{y}$$

$$\boldsymbol{\Omega} = \mathbf{V} \mathbf{Q}^{-1} \mathbf{V}^\top + \sigma_\varepsilon^2 \mathbf{R} \quad (\mathbf{V} \text{ maps matches to types})$$

$$\mathbf{Q} = \frac{1}{1 - \rho^2} \mathbf{S} \left[(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A} \right] \mathbf{S}, \quad \mathbf{S} = \text{diag}(\sigma_a^{-1} \mathbf{I}_{N_f}, \sigma_z^{-1} \mathbf{I}_{N_m})$$

$\mathbf{A}$ = adjacency, $\mathbf{D}$ = degrees, ρ on the edges, scales on the diagonal.

The precision matrix, one edge at a time

Zoom in on a single firm–executive pair (degree 1 each):

$$\mathbf{Q}_{\{i,m\}} = \frac{1}{1-\rho^2} \begin{bmatrix} \sigma_a^{-2} & -\rho \sigma_a^{-1} \sigma_z^{-1} \\ -\rho \sigma_a^{-1} \sigma_z^{-1} & \sigma_z^{-2} \end{bmatrix} \iff \text{Var} \begin{pmatrix} a_i \\ z_m \end{pmatrix} = \begin{bmatrix} \sigma_a^2 & \rho \sigma_a \sigma_z \\ \rho \sigma_a \sigma_z & \sigma_z^2 \end{bmatrix}$$

Off-diagonal entries of $\mathbf{Q}$ are nonzero **only along observed edges**.

On a forest, every linked pair has correlation exactly ρ .

Detour: Gaussian Markov random fields

What is a Gaussian Markov random field?

Jointly normal variables with a **sparse precision matrix**.

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Jointly normal variables with a **sparse precision matrix**.

Zero in $\mathbf{Q} \Leftrightarrow$ conditional independence given everyone else.

Conditional dependence is local — only direct neighbors.

Unconditional dependence travels along paths.

The best-known GMRF: an AR(1)

$x_t = \rho x_{t-1} + u_t$ on a line graph:

$$\mathbf{Q} \propto \begin{bmatrix} 1 & -\rho & & \\ -\rho & 1 + \rho^2 & -\rho & \\ & -\rho & 1 + \rho^2 & -\rho \\ & & -\rho & 1 \end{bmatrix}$$

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Partial correlation: only with neighbors. Unconditional autocorrelation at lag d : ρ^d .

Sparse precision, dense covariance.

On a tree, the GMRF looks just like an AR(1)

$$\text{Cov}(x_k, x_\ell) = \sigma_k \sigma_\ell \rho^{d(k,\ell)}$$

A tree is a branching timeline: unique paths, geometric decay per hop.

Our variance-stable $\mathbf{Q}$ keeps marginals at σ_a^2, σ_z^2 on any forest.

Real networks have cycles

Multiple paths push covariances above ρ^d .

The GMRF handles cycles exactly, through $\mathbf{Q}^{-1}$.

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But where does this particular $\mathbf{Q}$ *come from*?

Where the GMRF comes from: link formation

A statistical model of link formation

Not a competitive equilibrium — a selection model for which type pairs appear as links.

For equilibrium sorting with frictions, see (Shimer and Smith 2000, Eeckhout and Kircher 2010, Bagger and Lentz 2019)

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Not a competitive equilibrium — a selection model for which type pairs appear as links.

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Before links form, firm and executive types are **independent**:

$$a \sim \mathcal{N}(0, \omega_a^2), \quad z \sim \mathcal{N}(0, \omega_z^2)$$

Any correlation among *linked* pairs must come from link formation.

Match or not: a logit

Payoff from matching: the surplus $a_i z_m$.

Payoff from not matching: outside options, quadratic in own type.

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$$\Pr(L_{im} = 1 \mid a_i, z_m) = \frac{e^{\eta(a_i, z_m)}}{1 + e^{\eta(a_i, z_m)}}, \quad \eta = \kappa + \frac{1}{\tau} \left[a_i z_m - \alpha_1 a_i - \zeta_1 z_m - \frac{\alpha_2}{2} a_i^2 - \frac{\zeta_2}{2} z_m^2 \right]$$

τ scales match frictions. The cross derivative of η is positive: complementarity.

The likelihood of the observed graph

Joint density of types and links: prior $\times$ each edge that did or did not happen,

$$p(\mathbf{x}, E) \propto p_0(\mathbf{x}) \prod_{(i,m) \in E} \Pr(L_{im}=1 \mid \mathbf{x}) \prod_{(i,m) \notin E} \Pr(L_{im}=0 \mid \mathbf{x})$$

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Links are rare events. When e^η is small,

$$\Pr(L_{im} = 1 \mid a_i, z_m) \simeq e^{\eta(a_i, z_m)} \quad \text{and absent pairs drop out.}$$

The **sparse-link approximation**: keep only the positive links.

The tilted distribution is exactly our GMRF

Each observed link adds a quadratic term to the exponent:

$$p_E^+(\mathbf{x}) \propto p_0(\mathbf{x}) \prod_{(i,m) \in E} e^{\eta(a_i, z_m)} \propto \exp\left(-\frac{1}{2}\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{b}^\top \mathbf{x}\right)$$

$$\mathbf{Q} = \underbrace{\mathbf{Q}_0}_{\text{indep. prior}} + \underbrace{\frac{1}{\tau} \mathbf{C} \mathbf{D}}_{\text{repeated selection}} - \underbrace{\frac{1}{\tau} \mathbf{A}}_{\text{complementarity}} = \frac{1}{1 - \rho^2} \mathbf{S} \left[(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A} \right] \mathbf{S}$$

An equality, not an analogy: with $\tau = \sigma_a \sigma_z (1 - \rho^2) / \rho$, link formation *generates* the variance-stable precision.

When is the approximation valid?

The sparse-link approximation drops the shared logit denominators.

Valid when the graph is sparse and forest-like; dense cyclical regions invalidate it.

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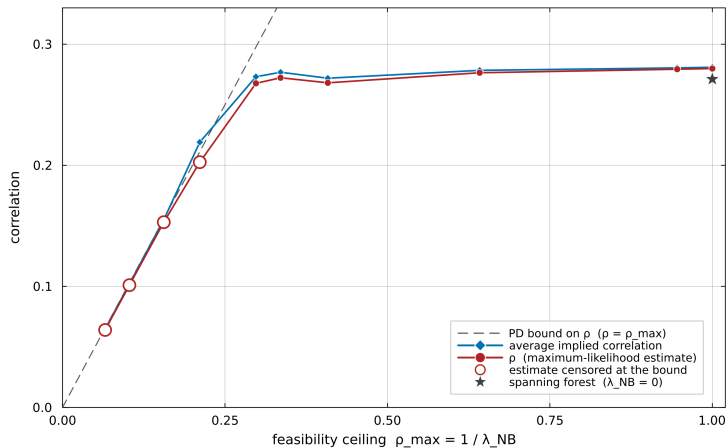
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Valid when the graph is sparse and forest-like; dense cyclical regions invalidate it.

So we **prune** the few cycle-closing edges in the dense core — keeping every node and every component — and estimate on the nearly-forest graph.

Opposite of leave-out: KSS lives on the cycles, we lean on the trees. But we keep the data.

Pruning: what the graph allows



Identification

Covariance decays along the tree

For two outcomes whose executives are h edges apart:

$$\text{Cov}(y_{i_1 m_1}, y_{i_2 m_2}) = \rho^{h-2} (\sigma_a + \rho \sigma_z)^2$$

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The **level** mixes σ_a , σ_z , ρ .

The **speed of decay** is ρ^2 per two hops — *nothing else*.

Four parameters, many covariance moments.

Gregory Clark measured decay on family trees

Correlation of social status between relatives d steps apart on the family tree (Clark 2023, PNAS):

$$\text{Corr}(y_i, y_j) = \theta^2 b^{d(i,j)}$$

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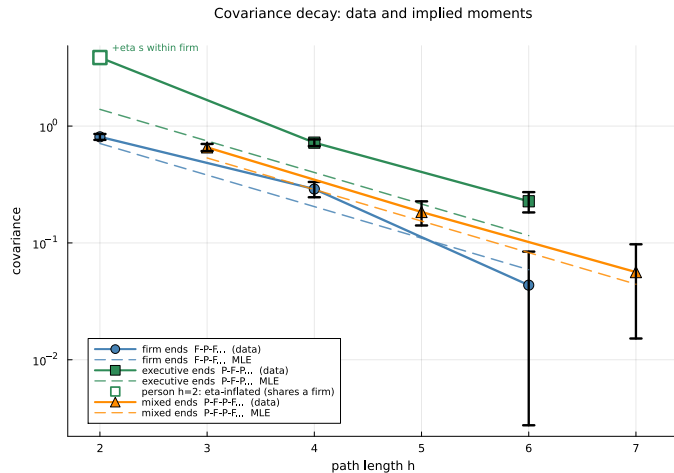
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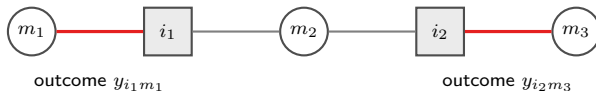
His formula is nearly ours. We got the idea from him.

Family trees are trees; mobility networks are almost trees.

Covariance decay in the Hungarian data

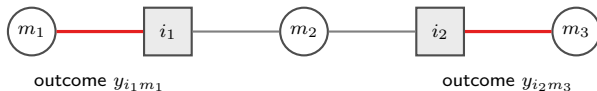


Prima facie evidence of sorting



m_1 and m_3 never share a firm; i_1 and i_2 never share an executive. **No common shocks.**

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m_1 and m_3 never share a firm; i_1 and i_2 never share an executive. **No common shocks.**

Yet their outcomes are correlated — and the model says by how much: $\rho^2(\sigma_a + \rho\sigma_z)^2$.

Ratios of such moments deliver ρ^2 by method of moments, before any likelihood.

What about within-firm shocks over time?

Consecutive spells at the *same firm* share transitory conditions: correlated ε , not sorting.

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Consecutive spells at the *same firm* share transitory conditions: correlated ε , not sorting.

But common shocks are **localized**. Sorting is identified by the **speed of decay over long chains**.

Three fixes that agree

- 1 **Minimum distance:** drop the contaminated same-firm moment
- 2 **Firm-average:** allow arbitrary within-firm dependence
- 3 **AR(1) match shocks:** estimate the error process (fifth parameter η)

All three give ρ between 0.52 and 0.55.

Results

Sorting is high: $\rho \approx 0.52\text{--}0.55$

Same graph, same outcomes (log real revenue, demeaned by year and industry):

Parameter	Naive MLE	Min. distance	Firm-average	AR(1) MLE
ρ	0.379	0.555	0.537	0.518
Implied corr. (a, z)	0.381	0.555	0.544	0.525

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The naive i.i.d.-error MLE is biased *down*: within-firm shocks masquerade as firm heterogeneity.

Preferred AR(1) fit: $\hat{\sigma}_a = 1.08$, $\hat{\sigma}_z = 0.24$, $\hat{\sigma}_\varepsilon = 1.72$, $\hat{\eta} = 0.51$.

GMRF vs leave-out on the common subgraph

Comparison is only possible on the sample both can use: the leave-out-connected block.

	KSS	GMRF (AR1)
Implied corr. (a, z)	TBD	TBD

Counterfactuals: what is sorting worth?

Expected revenue per match under the lognormal model:

$$\frac{Y}{n} = \exp \left\{ \mu_a + \mu_z + \mu_\varepsilon + \frac{1}{2} (V_a + V_z + V_\varepsilon + 2 \operatorname{Cov}(a, z)) \right\}$$

Hold the marginal distributions fixed, change the allocation, and read off the change in expected output.

Three counterfactuals

Scenario	Δ expected output
Random assignment ($\rho = 0$)	-12.9%
Compress talent dispersion, mean-preserving ($\sigma_z \rightarrow 0$)	-12.9%
Perfect sorting ($\rho = 1$)	$\approx +45\%$

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Perfect sorting ($\rho = 1$)	$\approx +45\%$

Killing talent dispersion costs exactly as much as scrambling the allocation: executives matter through *where they are*.

Perfect sorting is an upper bound, not a policy.

Threats to identification

The threat: similarity, not propagation

Outcomes on the network are correlated. Is it sorting?

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Outcomes on the network are correlated. Is it sorting?

The spells are at different times — this is *similarity*, not shock propagation.

Similarity could come from shared sector, region, or cohort: the homophily critique of network measurement.

(What we *measure* is homophily of latent types — the threat is homophily in something else.)

Our response: the Markov property is testable

The GMRF makes a strong, discrete prediction:

Conditional on your direct neighbors, the rest of the network is irrelevant.

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Conditional on your direct neighbors, the rest of the network is irrelevant.

An edge is a discrete event, not “being close”. A vague similarity story predicts second- and third-degree correlation *beyond* what direct links carry.

Check: does conditioning on direct links kill the residual correlation?

Later: controls for spatial and industry proximity (industry is already removed).

Placebo: disconnected components

Executives in disjoint components live in the same country, often the same industry.

The model predicts their outcomes have **zero** covariance.

Any common-shock story predicts otherwise.

Related work

Executive effects in the literature

Fenizia (2022): Italian public managers, KSS. Excellent — but retains only about 25% of managers. Exactly the problem we started from.

Metcalf, Sollaci and Syverson (2023): retail managers, *negative* estimated correlation.

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Metcalf, Sollaci and Syverson (2023): retail managers, *negative* estimated correlation.

Noise does that: inflated variances shrink the correlation, and the mechanical negative covariance of the two noisy effects pushes it below zero. (Mechanics in appendix.)

Our companion papers

Wohak and Koren (2025): leave-out on sparse graphs. KSS and the GMRF use *opposite* parts of the graph — cycles vs bridges. Even off the bridges, high leverage makes second-moment estimates noisy, and noise plus nonnegativity constraints means bias.

Koren, Orbán and Telegdy (2025): CEO value. A time-series estimator that removes spurious pre-trends from within-firm error correlation; after the correction, executive changes look unrelated to prior performance in the same Hungarian data.

Conclusion

Which estimator, when?

Network	Estimator
Dense (much mobility)	leave-out (KSS) — little data lost
Sparse, forest-like	GMRF
In between	run both, compare

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Network	Estimator
Dense (much mobility)	leave-out (KSS) — little data lost
Sparse, forest-like	GMRF
In between	run both, compare

Sparsity is where the GMRF is happy — and where fixed effects break.

The method travels

Sparse networks with latent node types are everywhere:

buyer–supplier networks, international trade, ...

The graph need not be bipartite.

Conclusion

- 1 Sorting between executive talent and firm fundamentals is **high**: $\rho \approx 0.52\text{--}0.55$
- 2 Four parameters replace millions of fixed effects — using **all** the data, not 10%
- 3 Identification is transparent: covariance decay along mobility chains, and it is testable
- 4 Random assignment would cost **12.9%** of revenue

Appendix

Acknowledgements



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European Research Council
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NEMZETI
KUTATÁSI, FEJLESZTÉSI
ÉS INNOVÁCIÓS HIVATAL

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Why fixed-effects correlations come out negative

The FE estimates are linear combinations of outcomes, including the noise.

Whenever the firm effect loads *positively* on a noise term, the executive effect at the same match loads *negatively*: the two dummies share the same variation.

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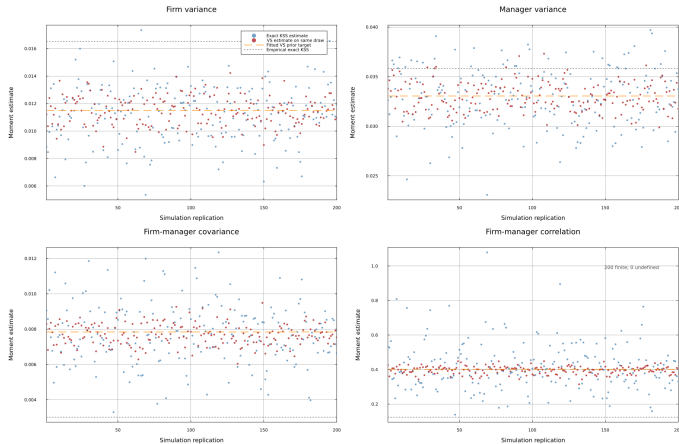
Whenever the firm effect loads *positively* on a noise term, the executive effect at the same match loads *negatively*: the two dummies share the same variation.

Inflated variances bias the correlation toward zero; the mechanical negative covariance pushes it further down.

Consistent with the literature: estimated correlations below what we find, some negative.

Leave-out is noisy even where it exists

Finite-sample dispersion of KSS and VS estimates on the common graph



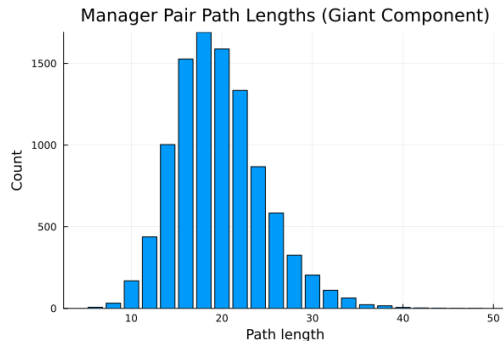
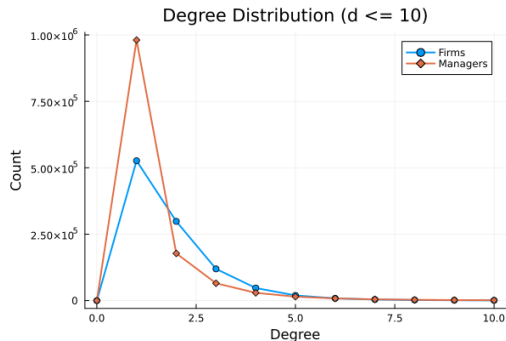
Simulations on the common graph: KSS second moments (blue) disperse far more than the GMRF (red).

Model fit: where log-revenue variance goes

Component	Share
Firm effects	28.4%
Executive effects	1.4%
Sorting covariance	6.6%
Match noise (within-firm correlated)	63.5%

The sorting covariance is several times the executive component.

The bipartite mobility graph



The giant component holds 42% of edges; the rest are mostly single firm–executive pairs.

Monte Carlo: correlated errors on the Hungarian graph

Hungarian common graph (K=91,085), 2000 reps/arm — truth $\rho = 0$

